

MODULE 03

SOFTWARE TESTING LIFE CYCLE (STLC) WITH AI

Online Shopping Website — Test Plan & Test Checklist

ACADEMIC ASSIGNMENT

Executive Summary

This assignment presents a structured quality-assurance approach for an Online Shopping Website using the Software Testing Life Cycle (STLC). The Test Plan defines objectives, scope, strategy, environment, test data, entry/exit criteria, AI-assisted activities, risks and deliverables. The Test Checklist provides 60 validation points across Login, Registration and Checkout—20 for each feature—with clear expected results. AI is positioned as a productivity assistant rather than a replacement for tester judgment: all AI-generated content must be reviewed, customized and approved before execution.

Q1. Test Plan for an Online Shopping Website

1. Test Objectives

- Verify that all critical e-commerce functions satisfy the specified requirements and business rules.
- Ensure users can register, authenticate, browse products, manage a cart, and complete checkout successfully.
- Validate positive, negative, boundary, error-handling, and security scenarios.
- Identify defects early and reduce the risk of production failures.
- Verify correct calculation of prices, discounts, shipping, tax, stock, and final order totals.
- Evaluate usability, performance, compatibility, accessibility, reliability, and basic application security.
- Use AI as an assistant for test-case generation, risk analysis, defect classification, and regression prioritization while keeping human review as the final authority.

2. Scope of Testing

Category	Coverage
In Scope	Registration, Login/Logout, Search, Product Details, Cart, Checkout, Payment sandbox flow, Order Confirmation, Order History/Tracking, validation, security, usability, compatibility, performance and accessibility.
Out of Scope	Internal implementation of external third-party systems and production infrastructure changes not directly related to application testing.

3. Major Modules to be Tested

Module	Coverage
User Account	Registration, login/logout, password validation,

	session management, profile and recovery.
Product & Search	Product listing, categories, search, filters, product details, price and stock.
Shopping Cart	Add/remove items, quantity updates, subtotal, stock validation and cart persistence.
Checkout	Address, shipping, coupon/discount, tax, order summary and total calculation.
Payment & Order	Payment success/failure handling, order creation, confirmation and order status.
Post-Purchase	Order history, tracking, cancellation/refund status where applicable.

4. Test Strategy

Testing Type	Approach
Functional Testing	Verify features against requirements using positive, negative, boundary and exploratory scenarios.
Integration Testing	Verify interactions among UI, backend/API, database, payment and order services.
System Testing	Validate complete end-to-end shopping journeys.
Regression Testing	Re-test fixed defects and affected critical areas after changes.
Security Testing	Check authentication, authorization, sessions, input validation and sensitive-data exposure.
Performance Testing	Check response time, stability and behavior under expected/concurrent load.
Compatibility Testing	Check supported browsers, operating systems, screen sizes and mobile/desktop layouts.
Usability & Accessibility	Check navigation, clarity, consistency, keyboard operation, labels and readable feedback.

5. Test Environment & Data

Area	Planned Configuration
Application	Test/Staging build of the Online Shopping Website
Browsers	Chrome, Edge, Firefox; Safari where supported
Devices	Desktop/laptop, tablet and smartphone
Database	Controlled test database with valid, invalid and boundary data
Network	Normal, slow and interrupted conditions where applicable

Accounts	New, existing, invalid and protected/locked scenarios
Payment	Sandbox/test payment environment only; no real payment credentials
Test Data	Valid/invalid users, stock levels, quantities, prices, coupons, addresses and payment outcomes

6. Entry & Exit Criteria

Criteria	Conditions
Entry Criteria	Approved requirements are available; test environment/build is ready; dependencies are accessible; test data and accounts are prepared.
Exit Criteria	Planned critical tests are executed; blocker/critical defects are resolved or formally accepted; regression is completed; test summary is prepared.

7. STLC Activities with AI Support

STLC Phase	Primary QA Activity	Responsible AI Assistance
Requirement Analysis	Identify testable acceptance criteria and ambiguous/missing requirements.	AI may highlight ambiguity and suggest questions; QA validates every suggestion.
Test Planning	Define scope, resources, schedule, risks, tools and entry/exit criteria.	AI can support risk brainstorming and prioritization; QA approves the final plan.
Test Case Design	Create scenarios, cases, data and expected results.	AI can draft positive/negative/boundary cases from requirements; testers review and customize them.
Environment Setup	Prepare build, database, browsers, devices, accounts and test data.	AI can generate setup/verification checklists; tester confirms the environment.
Test Execution	Execute cases, record evidence, actual results and pass/fail status.	AI can summarize evidence and help classify failures; testers determine actual results.
Defect Reporting	Log severity, priority, steps, expected/actual result and evidence.	AI can improve wording and identify possible duplicates; human verification is required.
Regression Testing	Re-test fixes and affected modules.	AI can prioritize regression cases using change impact

		and historical failures.
Test Closure	Summarize execution, defects, risks, lessons learned and release readiness.	AI can draft metrics and summaries; QA performs final review and sign-off.

8. Defect Severity & Priority Guide

Level	Meaning	Example
Critical	System/security failure with no practical workaround; blocks a critical business flow.	Checkout creates duplicate paid orders; severe authentication bypass.
High	Major functionality is broken and business impact is significant.	Valid users cannot log in; payment success is not recorded.
Medium	Important issue with a reasonable workaround.	Coupon calculation fails for one supported condition.
Low	Minor cosmetic or low-impact usability issue.	Alignment or wording issue with no functional impact.

9. Risk Assessment & Mitigation

Risk	Impact	Mitigation
Incorrect price/stock calculation	High	Use boundary, database and concurrency-focused tests; verify totals against independent calculations.
Payment integration failure	High	Use sandbox testing and cover success, failure, timeout, cancellation and retry scenarios.
Security vulnerability	High	Perform authentication, authorization, session and input-validation testing.
Browser/device incompatibility	Medium	Maintain a browser/device matrix and execute compatibility checks.
AI-generated inaccurate cases	Medium	Treat AI output as draft material; review, correct and approve before execution.
Late defect discovery	Medium	Prioritize critical user journeys and run continuous regression testing.

10. Deliverables

- Approved Test Plan
- Test Scenarios and Test Cases
- Login / Registration / Checkout Checklists
- Defect Reports with Evidence
- Test Execution & Regression Results

- Test Summary Report
- Release Readiness / Closure Recommendation

11. Requirement-to-Test Traceability (High-Level)

Requirement Area	Mapped Checklist IDs	Coverage
User Authentication	L-01 to L-20	UI, validation, security, session, logout and failure handling
User Registration	R-01 to R-20	Input validation, password policy, duplicate account, terms, database save
Shopping & Checkout	C-01 to C-20	Cart, stock, pricing, coupon, address, shipping, tax, payment and order data

Q2. Test Checklist

The following checklist is designed for practical execution. The tester should mark the Status column after validating the actual behavior and attach evidence (screenshot/log) when a case fails.

A. Login Checklist — 20 Validation Points

Test ID	Test Area	Validation Point	Expected Result	Status
L-01	Login UI	Email/username and password fields are displayed and clearly labeled.	Both required fields are visible and usable.	☐ Not Run
L-02	Login UI	Login/Sign In button is visible and actionable.	User can initiate authentication.	☐ Not Run
L-03	Valid Login	Registered email and correct password are submitted.	User is authenticated and redirected to the expected page.	☐ Not Run
L-04	Invalid Password	Valid email with incorrect password is submitted.	Authentication fails and a clear, safe error is displayed.	☐ Not Run
L-05	Unknown User	Unregistered email is submitted.	Login is rejected without revealing sensitive account information.	☐ Not Run
L-06	Empty Email	Email field is blank when login is submitted.	Required-field validation is displayed.	☐ Not Run

L-07	Empty Password	Password field is blank when login is submitted.	Required-field validation is displayed.	☐ Not Run
L-08	Both Empty	Both fields are blank and login is attempted.	Validation is shown and request is not accepted.	☐ Not Run
L-09	Email Format	Malformed email is entered.	Invalid-format validation prevents inappropriate submission.	☐ Not Run
L-10	Password Masking	Password is typed into the password field.	Password characters are masked.	☐ Not Run
L-11	Show/Hide Password	Password visibility control is used, if available.	Visibility changes correctly without changing the password value.	☐ Not Run
L-12	Email Policy	Email is entered with different casing where policy permits.	System follows the documented email case policy consistently.	☐ Not Run
L-13	Remember Me	Remember-me option is selected, if available.	Persistence follows the defined security policy.	☐ Not Run
L-14	Forgot Password	Forgot-password link is selected.	User enters the password-recovery flow.	☐ Not Run
L-15	Account Protection	Repeated failed attempts reach the configured threshold.	Configured lockout/rate-limit protection is triggered.	☐ Not Run
L-16	Session	Protected page is opened after successful login.	Authenticated session is maintained correctly.	☐ Not Run
L-17	Logout	User selects logout after authentication.	Session is terminated and user is returned to the public/login page.	☐ Not Run
L-18	Back Button	Browser Back is used after logout to access protected	Protected content is not available without re-authentication.	☐ Not Run

		content.		
L-19	Sensitive Data	Password is checked in URL, page text and error messages.	Password is not exposed.	☐ Not Run
L-20	Failure Handling	Login is attempted during a server/network failure.	Meaningful error is shown and application remains stable.	☐ Not Run

B. Registration Checklist — 20 Validation Points

Test ID	Test Area	Validation Point	Expected Result	Status
R-01	Registration UI	Registration form and required fields are displayed.	Fields are visible, labeled and usable.	☐ Not Run
R-02	Required Fields	Mandatory fields are left blank.	Required-field messages are shown.	☐ Not Run
R-03	Valid Registration	All required fields contain valid unique data.	Account is created successfully.	☐ Not Run
R-04	Duplicate Email	An already registered email is submitted.	Duplicate registration is prevented with a suitable message.	☐ Not Run
R-05	Email Format	Invalid email formats are entered.	Format validation rejects invalid input.	☐ Not Run
R-06	Password Minimum	Password below the defined minimum is entered.	Password policy message appears and submission is blocked.	☐ Not Run
R-07	Strong Password	Password satisfies all defined rules.	Password is accepted.	☐ Not Run
R-08	Confirm Mismatch	Password and confirmation do not match.	Mismatch validation is displayed and account is not created.	☐ Not Run
R-09	Confirm Match	Password and confirmation match.	Confirmation validation passes.	☐ Not Run
R-10	Name Validation	Name contains allowed	Name is accepted according to	☐ Not Run

		characters and length.	requirements.	
R-11	Phone Invalid	Invalid phone format or length is entered.	Invalid phone value is rejected.	☐ Not Run
R-12	Phone Valid	Valid phone number is entered.	Phone value is accepted.	☐ Not Run
R-13	Whitespace	Leading/trailing spaces are entered in text/email fields.	System trims or handles whitespace according to requirements.	☐ Not Run
R-14	Boundary Length	Minimum and maximum field lengths are tested.	Boundary values are accepted/rejected correctly.	☐ Not Run
R-15	Terms Unchecked	Mandatory Terms & Conditions checkbox is not selected.	Registration cannot proceed.	☐ Not Run
R-16	Terms Checked	Terms checkbox is selected.	Registration can proceed when other validations pass.	☐ Not Run
R-17	Password Masking	Password and confirmation fields are typed into.	Password values are masked.	☐ Not Run
R-18	Anti-Bot	CAPTCHA/anti-bot control is tested with invalid and valid challenges.	Invalid challenge blocks submission; valid challenge permits it.	☐ Not Run
R-19	Database Save	Successful registration is completed.	Exactly one correct user record is created.	☐ Not Run
R-20	Post Registration	Registration completes successfully.	Correct success/verification step and next navigation are shown.	☐ Not Run

C. Checkout Checklist — 20 Validation Points

Test ID	Test Area	Validation Point	Expected Result	Status
C-01	Cart	Checkout is	User is taken to	☐ Not Run

		initiated with at least one valid product.	checkout successfully.	
C-02	Empty Cart	Checkout is attempted with an empty cart.	Checkout is blocked or an appropriate message is shown.	☐ Not Run
C-03	Stock	Cart contains an unavailable/out-of-stock product.	System prevents an invalid purchase or requests correction.	☐ Not Run
C-04	Quantity Update	Quantity is increased within available stock.	Cart quantity and totals update correctly.	☐ Not Run
C-05	Quantity Boundary	Quantity is zero, negative or above allowed stock.	Invalid quantity is rejected or corrected.	☐ Not Run
C-06	Item Price	Product price is displayed at checkout.	Price matches the applicable current cart/product price.	☐ Not Run
C-07	Subtotal	Multiple products are in the cart.	Subtotal equals sum of item price × quantity.	☐ Not Run
C-08	Valid Coupon	A valid eligible coupon is applied.	Correct discount is applied.	☐ Not Run
C-09	Invalid Coupon	Invalid or expired coupon is entered.	Coupon is rejected and no incorrect discount is applied.	☐ Not Run
C-10	Address Valid	All required shipping-address fields contain valid values.	Address is accepted.	☐ Not Run
C-11	Address Required	A mandatory address field is blank.	Checkout cannot continue and validation is shown.	☐ Not Run
C-12	Shipping Method	A shipping method is selected.	Correct shipping charge is reflected.	☐ Not Run
C-13	Tax	Tax applies	Correct tax is	☐ Not Run

		according to configured rules.	calculated and displayed.	
C-14	Final Total	Subtotal, discount, tax and shipping are combined.	Final payable amount is mathematically correct.	☐ Not Run
C-15	Payment Method	Available payment methods are displayed.	Eligible payment method can be selected.	☐ Not Run
C-16	Payment Success	Successful sandbox payment is completed.	Order is created and marked paid/successful correctly.	☐ Not Run
C-17	Payment Failure	Payment is declined, fails or times out.	Order is not incorrectly marked paid and failure is handled clearly.	☐ Not Run
C-18	Duplicate Submit	Place Order/Pay is clicked repeatedly.	Duplicate order creation is prevented.	☐ Not Run
C-19	Confirmation	Successful order placement is completed.	Unique order ID and confirmation details are displayed/sent.	☐ Not Run
C-20	Order Data	Completed order is checked in order history/database.	Items, quantities, amount, address and status are stored correctly.	☐ Not Run

12. Overall Acceptance Criteria

- All critical Login, Registration and Checkout scenarios pass.
- No blocker or critical defect remains open at release time unless formally accepted.
- High-severity defects are fixed or have documented risk acceptance.
- Authentication, session handling and sensitive-data protection meet the defined security requirements.
- Cart, stock, discount, shipping, tax, payment and order totals are accurate.
- Supported browsers and device layouts work within the agreed compatibility matrix.
- AI-assisted test artifacts have been reviewed and customized by the QA team.

13. Conclusion

This test plan and checklist provide a complete, practical and measurable STLC-based quality-assurance framework for an Online Shopping Website. By combining structured manual validation with carefully reviewed AI assistance, the testing process can become faster without sacrificing accuracy or accountability. The 60 validation points focus on the highest-value customer

journeys—authentication, account creation and checkout—while the overall plan addresses integration, regression, security, performance, compatibility, usability and accessibility. Successful completion of these activities provides strong evidence for release readiness and helps reduce the risk of defects reaching end users.